



Psychological reactance to system-level policies before and after their implementation

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Governments need to develop and implement effective policies to address pressing societal problems of our time, such as climate change and global pandemics. While some policies focus on changing individual thoughts and behaviors (e.g., informational interventions, behavioral nudges), others involve systemic changes (e.g., car bans, vaccination mandates). Policymakers may use system-level policies to achieve socially desirable outcomes, yet often refrain from doing so because they anticipate public opposition. In this article, we propose that people's psychological reactance driving this opposition is a transient phenomenon that dissipates once system-level policies are in place. Using secondary survey data ($N = 49,674$) and experimental data (six studies; $N = 4,629$; all preregistered), we document that psychological reactance to system-level policies is greater when they are planned (ex ante implementation) than when they are already implemented (ex post implementation). We further demonstrate that this effect can be observed across various intervention contexts and provide insights into its underlying psychological mechanisms. Specifically, ex ante vs. ex post the system-level policy's implementation, individuals focus more on the transition-induced personal losses than on the prospective societal outcome gains. In line with this perspective, we show that the decline in reactance to system-level policies after their implementation is mediated and moderated by the salience of personal losses, and that the initial reactance to such policies is mitigated by the salience of societal gains. These findings suggest that the public's negative reactions to system-level policies are more transient than previously thought and can help policymakers design effective interventions.

psychological reactance | system-level policies | mandates | public policy

To successfully address some of the most significant societal challenges of our time, such as climate change or global pandemics, large-scale behavioral changes are necessary. Hence, there is an urgent need to develop and implement effective policies that support widespread behavior change (1, 2). To this end, policymakers can implement policies that focus on changing individual thoughts and behaviors—that is, individual-level or i-frame policies, such as subtle nudges, information provision, or education. Alternatively, policymakers can implement policies aimed at systemic changes—that is, system-level or s-frame policies, which often impose costs or restrictions on people. For instance, policymakers may increase gasoline tax or even restrict car use in cities to reduce air pollution (3). Another example is the increase of tobacco taxation or the implementation of smoking bans to decrease the prevalence of smoking-related diseases (4). While such system-level policies may benefit society as a whole now or in the future, many academics and experts have prioritized individual-level policies in the past (1). This is partly due to concerns that system-level policies, by directly or indirectly limiting people's individual autonomy, may trigger psychological reactance. According to reactance theory, policies that eliminate individual choices and freedoms create a state of psychological reactance (5) reflected in negative cognitions, emotions, and behaviors toward the policy and policymakers (5–10), which may in turn lower their public acceptance (11).

Consistent with this view, government initiatives to implement systemic changes often provoke heated public debates and opposition (12–14). For example, the introduction of mandatory seatbelt legislation faced initial resistance from citizens (15). Similarly, the implementation of stricter smoking laws over the last two decades was met with fierce opposition from many citizens (16, 17). Yet, opposition against such systemic changes has seemingly disappeared once they have been implemented. This observation is also in line with psychological research suggesting that individuals have a general tendency to prefer the status quo and often underestimate their ability to emotionally adjust to changes (e.g., ref. 18). Consequently, resistance to system-level policies might not reflect actual preferences against the policy and its consequences but rather a temporary psychological

Significance

Policymakers have the capacity to implement system-level policies that may impose costs or restrictions on people to achieve socially desirable outcomes (e.g., mandatory vaccinations). However, a common concern with system-level policies is their potential to cause psychological reactance, undermining public acceptability and support. We study the temporal dynamics of reactance to system-level policies. Using both survey and experimental data, we provide causal evidence that psychological reactance to system-level policies is greater during the planning phase than after their implementation. We offer an explanation for why this effect occurs and propose strategies to mitigate psychological reactance to system-level policies, which can help policymakers design and build support for such policies if deemed necessary for tackling societal challenges.

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reaction that is particularly intense before implementation of the policies and diminishes afterward. Despite its relevance for policymakers and humanity's ability to solve critical societal challenges, empirical support for this hypothesis remains limited and inconclusive. For instance, the current body of evidence, primarily based on correlational analyses of survey data within the health domain, shows that the introduction of some system-level policies is associated with an increase in public support for these policies, whereas the introduction of others is not (7). Moreover, the existing evidence for this variance between pre- and postimplementation periods lacks causal clarity and fails to explore the underlying psychological mechanisms. Thus, there is a need for an improved and broader understanding of this effect (14). Here, we aim to close this gap.

We systematically investigate the temporal dynamics of psychological reactance to restrictive system-level policies. Our research is guided by a previously underexplored theoretical perspective on why psychological reactance should be more pronounced during the preimplementation phase of a restrictive policy (*ex ante*) than after its implementation (*ex post*). By moving beyond correlational evidence, our research seeks to provide a deeper understanding of this effect's underlying mechanisms and its implications for addressing and mitigating psychological reactance to system-level policies.

In a first step, relying on large-scale secondary survey data from 26 European countries, we analyzed how opposition to smoking bans differed across countries that implemented these bans at different times vs. those that did not implement bans at all. This allowed us to employ a difference-in-differences (DID) approach to compare people's opposition to a smoking ban, controlling for confounding variables. In a second step, we provide additional causal evidence for the proposed effect and its psychological boundary conditions through three (hypothetical) survey experiments. In these experiments, we assessed the generalizability of this phenomenon across a broad spectrum of policy contexts crucial for addressing major societal challenges, such as combating climate change (e.g., reducing car usage; ref. 19), managing global pandemics (e.g., mandating vaccination; ref. 20), and enhancing public health (e.g., reducing alcohol consumption; ref. 21). Finally, we tested our theoretical framework in three additional experimental studies, which included theory-informed interventions designed to mitigate psychological reactance from the outset, controlling for alternative explanations (e.g., emotional adaptation).

Theoretical Background. Why should psychological reactance be stronger before rather than after the implementation of a system-level policy when individuals have actually experienced the policy's consequences? We propose that this effect can be understood through the lens of a transition heuristic framework (22). When individuals anticipate moving from their current state (A) to a new state (B), they are more inclined to evaluate the process of transition between the two states (e.g., *giving up A* in exchange for B) rather than evaluating the states themselves (22). However, once the transition has occurred, people are more inclined to evaluate the new state (e.g., the value of B). This transition heuristic account therefore implies a shift from focusing on evaluating change-of-states to evaluating states-in-themselves. Consequently, individuals who are generally loss-averse may not focus on the potential benefits or drawbacks of the new state (B) but on what they will lose by making the transition (22–25). This framework leads us to theorize that before vs. after a system-level policy is implemented, individuals perceive it as a transition, focusing more on the *losses* than on the actual outcome of the policy. System-level policies are typically designed to benefit

society as a whole, albeit at the expense of individual choices and freedoms (7, 26). Consequently, before a system-level policy is implemented (i.e., *ex ante* its implementation), individuals should focus more on intervention-induced losses at the individual level (e.g., convenience, personal freedom), which may induce psychological reactance. In contrast, after a system-level policy has been implemented (i.e., *ex post* its implementation), individuals should be less likely to evaluate the intervention as a transition that causes personal losses (the transition has already occurred). Instead, people should be more likely to focus on the societal gains of the novel policy (increase in overall welfare), which, in turn, may reduce psychological reactance.

Results

Before testing the psychological processes as proposed by our theoretical framework, we first aim to establish evidence for the effect of interest: a diminishing opposition to system-level policies after vs. before their implementation. To this end, Study 1 (total $N = 49,674$) used secondary data from 26 countries on public opposition to smoking bans in indoor workplaces from two representative surveys conducted in the European Union in 2005 and 2006 (27, 28). In between these data collection waves, three countries (Belgium, Scotland, and Spain) implemented a smoking ban in indoor workplaces (intervention group; $n = 4,280$). Belgium implemented a ban on smoking in all workplaces in January 2006; Scotland implemented a smoking ban in all workplaces and public spaces such as hospitality venues in March 2006; and Spain implemented a smoking ban that prohibited smoking in workplaces and public spaces in January 2006, but also incorporated exceptions that allowed a considerable portion of hospitality venues to decide whether or not to implement the ban (29, 30). All other 23 countries implemented no change in their smoking policy (control group; $n = 45,394$). We employed a difference-in-differences (DID) approach to compare people's opposition to a smoking ban in indoor workplaces (1 = Totally in favor to 4 = Totally opposed) in Belgium, Scotland, and Spain before and after the implementation with that of countries that did not plan and implement a smoking ban in the relevant period.

Because of the varying scope and timing of smoking bans across the countries in the intervention group, we sought to account for the country-level heterogeneity in the treatment effect (31, 32). To this end, we included interactions between each intervention country and the year of policy implementation (31). Results from this OLS regression model show that the public was more strongly opposed to a smoking ban in countries where the policy was planned (vs. not planned) for the coming year (main effect: unstandardized $b_{\text{Belgium}} = 0.094$, $SE = 0.04$, $P < 0.05$; $b_{\text{Scotland}} = 0.178$, $SE = 0.04$, $P < 0.001$; $b_{\text{Spain}} = 0.158$, $SE = 0.04$, $P < 0.01$; Table 1). In both Belgium and Scotland, this public opposition significantly decreased following the implementation of the policy (interaction effects: $b_{\text{Belgium}} = -0.102$, $SE = 0.02$, $P < 0.001$; $b_{\text{Scotland}} = -0.368$, $SE = 0.02$, $P < 0.001$). In Spain, however, we saw no significant decrease after the implementation of the policy (interaction effect: $b_{\text{Spain}} = 0.044$, $SE = 0.02$, $P = 0.078$; Fig. 1). This pattern remained consistent when we estimated a series of separate models for Belgium, Scotland, and Spain, which included additional control variables related to smoking-related behaviors, sociodemographic characteristics, political orientation, and citizen's country and occupation (*SI Appendix*). As an additional robustness check, we re-estimated the models using doubly robust DID estimators, following established guidelines (33). These estimators enforce parallel trends conditional on the control variables before the intervention, thus enhancing robustness to violations of the

Table 1. Regression model of public opposition to system-level policies, i.e., smoking ban in indoor workplaces (Study 1)

Independent variable*	Dependent variable: Public opposition to system-level policy
Implementation period	−0.080*** (0.024)
Belgium	0.094** (0.044)
Implementation period × Belgium	−0.102*** (0.024)
Scotland	0.178*** (0.044)
Implementation period × Scotland	−0.368*** (0.024)
Spain	0.158*** (0.044)
Implementation period × Spain	0.044* (0.024)
Constant	1.469*** (0.044)
Observations	49,064
R-squared	0.005

Notes: Coefficients represent unstandardized regression coefficients. SE are clustered at the country level and shown in parentheses.
*Countries of the intervention group (i.e., Belgium, Scotland, and Spain) included as individual dummy variables, thus, each country-specific coefficient reflects the difference in opposition between the respective intervention country and the countries of the control group before the implementation of the policy. *** $P < 0.01$, ** $P < 0.05$, * $P < 0.1$.

parallel trends assumption (34). This analysis corroborated our main results: Belgium and Scotland showed significant decreases in public opposition after the implementation of the policy ($b_{\text{Belgium}} = -0.084$, $SE = 0.02$, $P < 0.001$; $b_{\text{Scotland}} = -0.205$, $SE = 0.02$, $P < 0.001$), while Spain did not ($b_{\text{Spain}} = 0.020$, $SE = 0.02$, $P = 0.303$). Unlike Belgium and Scotland, Spain implemented only a partial ban on smoking with a more permissive

smoking legislation (30). This led to heated debates and proved less effective than the comprehensive ban on public smoking that eventually replaced it (29, 35). We therefore speculate that this partial smoking ban, which included more loopholes than the smoking bans in Belgium and Scotland (30), may have contributed to the sustained public opposition even after its implementation.

To further strengthen the findings of Study 1, we considered secondary survey data on public opposition to system-level policies in two other domains: the introduction of mandatory seatbelt legislation in the state of New York (36) and the introduction of a speed limit of 100 km/h (approx. 62 mph) on motorways between 6:00 a.m. and 7:00 p.m. in the Netherlands (37). These data provided further suggestive evidence that opposition to these two system-level policies decreased after their implementation (see [SI Appendix](#) for details).

In sum, the results of Study 1, together with these additional findings, provide initial support for our hypothesis that psychological reactance to system-level policies is greater before than after the policy’s implementation. While these findings offer suggestive evidence for our proposed effect in real-world settings with high external validity, they do not permit strong causal conclusions as the study relies on two time points only, which limits our ability to rigorously test the parallel trends assumption. To address this, the subsequent studies aim to provide further and causal evidence for our hypothesis and the proposed underlying process by utilizing a fully randomized experimental setting with high internal validity.

Study 2 ($N = 719$ UK participants) aimed to test our main effect hypothesis and its generalizability across different system-level policy contexts. Moreover, we measured psychological (state) reactance more directly. Participants were assigned to one of six conditions in a 2 (policy implementation stage: ex ante vs. ex post) × 3 (policy context: cars vs. alcohol vs. meat) between-participants design. Specifically, participants read that the government plans to increase (vs. increased) taxes in 1 y from now (vs. 1 y ago) on cars, alcohol, or meat. We measured participants’ psychological reactance by assessing their anger in response to the system-level policy on a six-point Likert-type scale.

An ANOVA revealed a significant main effect of ex ante vs. ex post policy interventions on reactance ($M_{\text{ex ante}} = 4.20$ vs. $M_{\text{ex post}} = 3.46$; $F(1, 713) = 40.60$, $P < 0.001$, $\eta^2 = 0.05$, 95% CI [0.03, 0.09]). We did not find a significant interaction effect between ex ante vs. ex post policy interventions and policy context

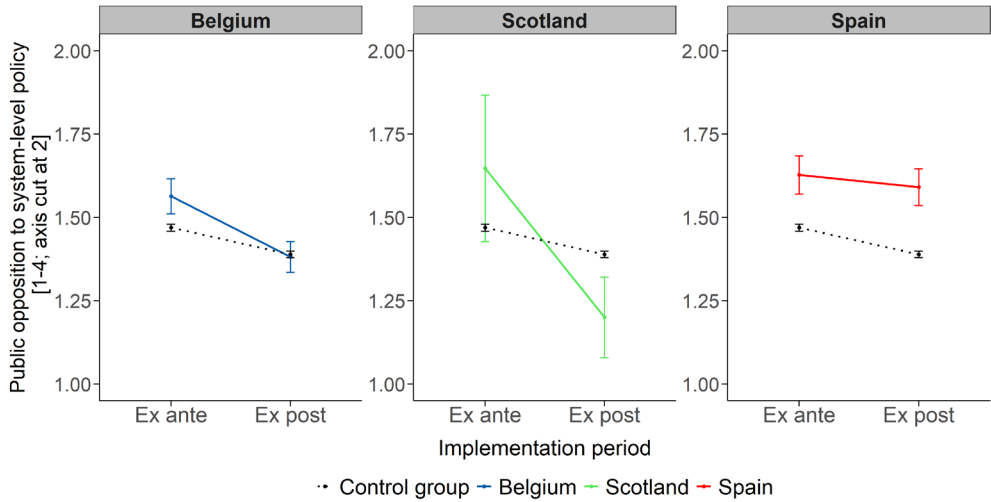


Fig. 1. Public opposition to the system-level policy when it was planned (ex ante implementation) and when it was implemented (ex post implementation) in the three countries that implemented the policy vs. those that did not plan and implement the policy in Study 1 ($N = 49,674$; $M_{\text{age}} = 47.81$, $SD_{\text{age}} = 18.43$). Colored (black) dots and whiskers indicate the observed means and their 95% CI in the countries of the intervention (control) group.

($F(2, 713) = 0.64, P = 0.527$). Across all three policy contexts, participants' reactance to system-level policies was greater ex ante than ex post their implementation (all P s < 0.01; $d_{\text{cars}} = 0.58$, 95% CI [0.32, 0.84]; $d_{\text{alcohol}} = 0.50$, 95% CI [0.24, 0.75]; $d_{\text{meat}} = 0.35$, 95% CI [0.10, 0.61]; see Fig. 2 for an overview). These findings thus provide causal evidence for the hypothesized effect across several policy contexts.

Studies 3A ($N = 244$ UK participants) and 3B ($N = 809$ German participants) investigated ex ante vs. ex post psychological reactance to the implementation of mandatory vaccination and a speed limit on German motorways, respectively. Both of these system-level policies have caused public debates (13, 38). Additionally, both studies explored whether participants' general attitudes toward these policies moderate the potential decline in reactance to system-level policies after their implementation. We argued that reactance to system-level policies represents a transient phenomenon rather than stable individual preferences against them. If so, then people's reactance to these policies should be greater before vs. after the policy's implementation regardless of whether people generally support the policy in the first place. Both studies used a between-participants design. In Study 3A, participants read that a new disease was currently spreading across many different geographical areas, for which a vaccine was available that has been shown to effectively protect against the disease. In the ex ante (vs. ex post) system-level policy intervention condition, participants read that to limit the spread of the disease, the government plans to implement (vs. implemented) a vaccination mandate to get vaccinated in 1 y from now (vs. 1 y ago). In Study 3B, participants read that to reduce carbon emissions, the government plans to introduce (vs. introduced) a speed limit of 120 km/h (approx. 75 mph) on German motorways in 1 y from now (vs. 1 y ago). As the dependent variable, we measured psychological reactance to the system-level policy in both studies using the same question as in Study 2. As a potential moderator variable, we additionally measured participants' general attitudes toward the

policy ("In general, I have a positive attitude towards vaccinations"/"I am in favor of a speed limit on German motorways"; 0 = Very much disagree to 100 = Very much agree/0 = Not at all to 100 = Very much).

Consistent with our previous findings, in both studies participants' psychological reactance to the mandate was greater ex ante than ex post its implementation (implementation of mandatory vaccination: $M_{\text{ex ante}} = 3.29$, $SD = 1.72$ vs. $M_{\text{ex post}} = 2.71$, $SD = 1.61$; $t(242) = 2.69, P < 0.01, d = 0.34$, 95% CI [0.09, 0.60]; implementation of a speed limit: $M_{\text{ex ante}} = 2.80$, $SD = 1.84$ vs. $M_{\text{ex post}} = 2.39$, $SD = 1.73$; $t(807) = 3.29, P < 0.01, d = 0.23$, 95% CI [0.09, 0.37]; Fig. 2). Moreover, in both studies, we found no evidence for a significant moderation effect by general attitudes toward the policy ($b_{\text{vaccination}} = -0.014, P = 0.156$; $b_{\text{speed limit}} = 0.002, P = 0.526$). This absence of a meaningful moderation effect suggests that the decline in ex ante vs. ex post reactance to system-level policies represents a psychological response that is independent of whether individuals generally favor or oppose the policy.

Studies 4A ($N = 600$ UK participants) and 4B ($N = 1,635$ UK participants) aimed to test our proposed transition heuristic account (22) by providing mediation and moderation evidence. Based on this account, we argued that people's reactance to system-level policies is greater ex ante vs. ex post their implementation because individuals perceive the policy as a transition and, thus, focus more strongly on transition-induced personal losses than societal gains. In Study 4A, we tested this explanation using the context of banning cars for commuting to work. In the ex ante (vs. ex post) condition, participants read that the government plans to forbid (vs. forbade) people to commute to work by car in 1 y from now (vs. 1 y ago). As the dependent variable, we measured participants' psychological reactance to this policy using three items adapted from previous literature (e.g., "I would feel that the government restricts my personal freedom"; 1 = Not at all to 6 = Very much; Cronbach's $\alpha = 0.93$). As potential process

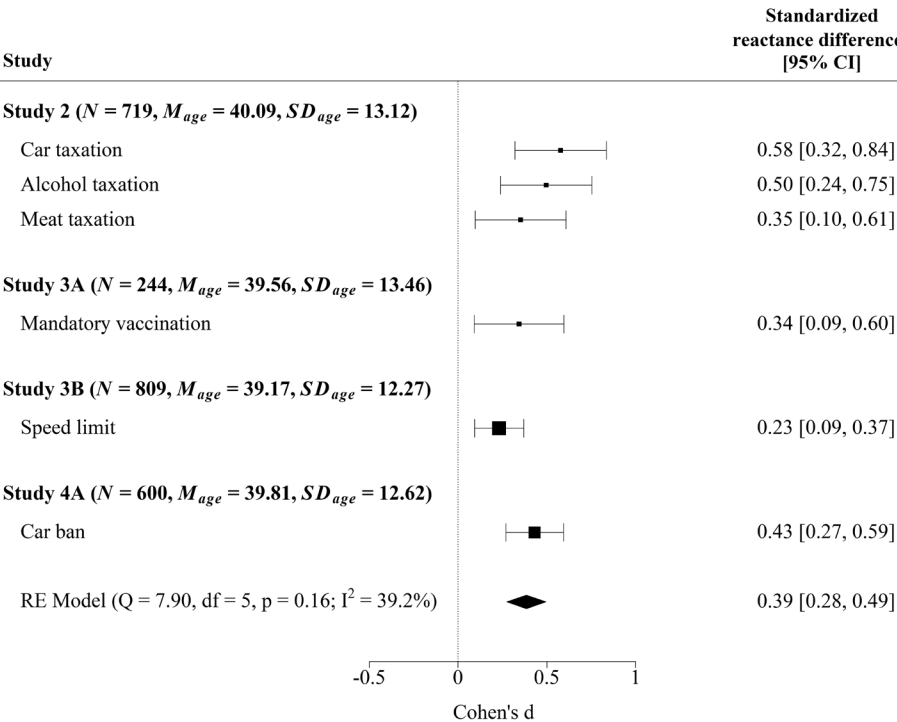


Fig. 2. Forest plot of mean differences in psychological reactance to system-level policies before (ex ante) vs. after (ex post) their implementation across Studies 2–4A. Note: Differences in psychological reactance refer to standardized differences. The experimental materials were comparable and varied in policy contexts as described in *Materials and Methods*; thus, the four studies can be understood as conceptual replications. The overall effects were calculated using a random effects model for meta-analysis. Q and I^2 were used for heterogeneity assessment among the studies. CIs refer to 95% CI.

variables, we measured personal loss perceptions (“I think about what I personally lose from this decision”; 1 = Strongly disagree to 6 = Strongly agree), societal gain perceptions (“I think about what the society gains from this decision”; 1 = Strongly disagree to 6 = Strongly agree), and the relative focus on personal vs. societal consequences (“I focus more on how this decision impacts”; 1 = My personal life to 6 = The whole society).

Consistent with our theorizing, a multiple mediation model revealed that all of the latter three measures can explain why participants’ reactance to the intervention is greater ex ante than ex post the implementation of the policy, but personal loss perceptions had the strongest mediation effect (indirect effect: $b = 0.281$, $SE = 0.07$, 95% CI [0.14, 0.43]). Participants indicated experiencing greater personal losses ex ante than ex post the implementation of the system-level policy ($M_{\text{ex ante}} = 4.70$, $SD = 1.26$ vs. $M_{\text{ex post}} = 4.29$, $SD = 1.37$; $t(598) = 3.84$, $P < 0.001$, $d = 0.31$, 95% CI [0.15, 0.47]). This, in turn, was associated with greater reactance to the policy ($b = 0.669$, $SE = 0.04$, $P < 0.001$; Fig. 2). A sensitivity analysis (39) demonstrated that the mediation effect of personal loss perceptions appears relatively robust to unobserved confounders (sensitivity parameter $\rho = 0.70$; ρ ranges from -1 to $+1$; larger absolute values of ρ indicate greater robustness to unmeasured confounders). In an additional Study S1, reported in *SI Appendix*, we replicate these findings controlling for related processes including the extent to which people think they can change the implementation of the system-level policy and the level of adaptation to the system-level policy (18, 40).

The mediation analysis of Study 4A relies on several implicit assumptions and, therefore, the causal effect should be interpreted with caution (41). Study 4B sought to provide a direct causal test of whether personal loss perceptions underlie increased ex ante vs. ex post psychological reactance to system-level policies through experimental manipulation of the proposed mediator (42). Specifically, we manipulated loss frame salience by asking participants to take a perspective that focuses on the losses caused by this policy and further asked them to provide a short qualitative description of such losses. The experiment thus applied a 2 (implementation stage: ex ante vs. ex post) $\times$ 2 (salience: loss frame vs. control) between-participants design. We used the same intervention context (i.e., banning cars for commuting to work) and psychological reactance measure (Cronbach’s $\alpha = 0.91$) as in Study 4A.

An ANOVA revealed a significant main effect of ex ante vs. ex post system-level policy on psychological reactance ($F(1, 1631) = 37.14$, $P < 0.001$, $\eta^2 = 0.02$, 95% CI [0.01, 0.04]) and a significant main effect of loss frame salience vs. control ($F(1, 1631) = 29.52$, $P < 0.001$, $\eta^2 = 0.02$, 95% CI [0.01, 0.03]). Importantly, as pre-registered, the analysis also yielded a significant interaction effect ($F(1, 1631) = 7.22$, $P < 0.01$, $\eta^2 = 0.01$, 95% CI [0.00, 0.01]). In the control condition, participants indicated a substantially larger reactance to the planned system-level policy compared to the policy that had been already implemented ($M_{\text{control, ex ante}} = 4.60$, $SD = 1.44$ vs. $M_{\text{control, ex post}} = 4.00$, $SD = 1.47$; $F(1, 1631) = 39.75$, $P < 0.001$, $d = 0.41$, 95% CI [0.27, 0.55]), replicating our previous findings. However, after inducing a loss frame in both conditions, the difference in reactance to a planned vs. implemented system-level policy was significantly reduced ($M_{\text{loss frame, ex ante}} = 4.79$, $SD = 1.27$ vs. $M_{\text{loss frame, ex post}} = 4.55$, $SD = 1.29$; $F(1, 1631) = 5.64$, $P = 0.018$, $d = 0.18$, 95% CI [0.04, 0.32]). While the observed interaction effect is small in magnitude, it supports our reasoning that people are in a “natural” loss frame when a restrictive policy is planned vs. implemented (Fig. 3). That is, when the system-level policy was planned (ex ante), inducing a loss frame had a small and only marginally significant effect on reactance compared to the control condition ($F(1, 1631) = 3.76$, $P = 0.053$, $d = 0.14$,

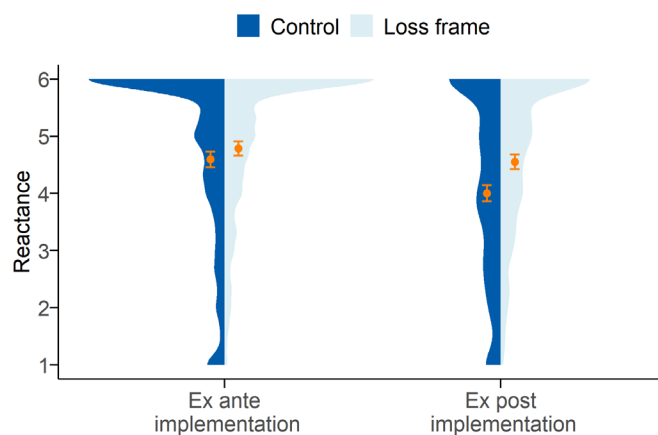


Fig. 3. Effect of loss frame salience on psychological reactance to the system-level policy across implementation stage conditions in Study 4B ($N = 1,635$; $M_{\text{age}} = 39.85$, $SD_{\text{age}} = 13.34$). Colored areas represent rotated kernel density distributions of individual responses. Orange dots and whiskers indicate the observed means and their 95% CI.

95% CI [0.00, 0.27]). When the system-level policy was implemented (ex post), however, this difference was significantly more pronounced ($F(1, 1631) = 33.07$, $P < 0.001$, $d = 0.40$, 95% CI [0.26, 0.54]). In sum, these findings support our theorizing: when a restrictive policy is planned vs. implemented, people more strongly focus on the policy-induced losses.

With Study 4C ($N = 622$ UK participants), we aimed to provide further evidence for the transition heuristic account and also tested a potential intervention to reduce ex ante reactance to system-level policies. We argued that when a system-level policy is planned (i.e., ex ante its implementation), people focus more naturally on what they might lose personally rather than on the benefits the policy might bring to society at large. If correct, then an intervention that communicates the societal gains of a system-level policy—and thus makes these gains more salient—should help reduce ex ante reactance (43). To test this prediction, participants were assigned to one of two conditions (salience: societal benefits vs. control) in a between-participants design with the same policy context (i.e., banning cars for commuting to work) and psychological reactance measure ($\alpha = 0.93$) as in Studies 4A and 4B. All participants read that the government plans to forbid people to commute to work by car in 1 y from now (i.e., only information about the ex ante policy intervention was provided). However, in the societal benefits condition, participants were additionally informed that the government cited benefits for society at large as the main reason for its decision, such as improving public health, addressing climate change, and making streets safer (*Materials and Methods*).

As predicted, when participants were informed that the government plans to implement the system-level policy because of its benefits for society at large, psychological reactance to the policy decreased significantly ($M_{\text{societal benefits}} = 3.99$, $SD = 1.58$ vs. $M_{\text{control}} = 4.40$, $SD = 1.48$; $t(620) = -3.31$, $P < 0.01$, $d = -0.27$, 95% CI [−0.42, −0.11]; Fig. 4). This finding further supports our theoretical account and provides a useful intervention to reduce opposition against planned system-level policies. That is, when the government plans a system-level policy, and when people are made aware of the policy’s broader benefits, they are less likely to feel reactance to it.

Discussion

Policymakers can address pressing societal problems such as climate change or global pandemics both through individual-level policies (i.e., encouraging individual behavior change) and

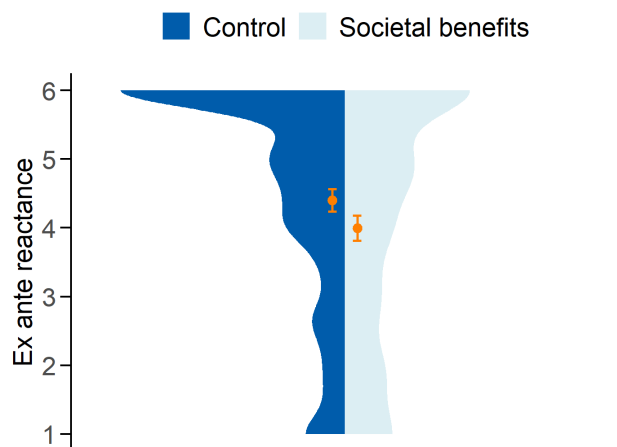


Fig. 4. Effect of societal benefits salience on psychological reactance ex ante the implementation of the system-level policy in Study 4C ($N = 622$; $M_{\text{age}} = 40.61$, $SD_{\text{age}} = 13.57$). Colored areas represent rotated kernel density distributions of individual responses. Orange dots and whiskers indicate the observed means and their 95% CI.

system-level policies (i.e., implementing systemic changes) (1). System-level policies (e.g., taxes or mandates) have the potential to elicit large-scale behavioral change and implement outcomes that benefit society at large, now or in the future. However, many experts and scholars raise concerns over system-level policies and often favor individual-level policies (e.g., nudges, education campaigns) because there are reasonable and evidence-based concerns that system-level policies may lead to psychological reactance, which may, in turn, lower their public acceptance (11). It is, therefore, no surprise that policymakers often hesitate to implement system-level policies because they are concerned about people's resistance once they are implemented.

In seven studies, using secondary survey data, experimental data, and different policy contexts, we consistently and causally demonstrate that psychological reactance to system-level policies decreases after their implementation. Our secondary survey data from 26 European countries ($N = 49,674$; $M_{\text{age}} = 47.81$, $SD_{\text{age}} = 18.43$) provided suggestive evidence for this decrease in real-world settings (DID effects: $b_{\text{Belgium}} = -0.102$, $P < 0.001$, $b_{\text{Scotland}} = -0.368$, $P < 0.001$, corresponding to an approximate 6% and 17% decrease after the policy implementation; see Fig. 1 and *SI Appendix* for details). Our experimental data ($N = 4,629$; $M_{\text{age}} = 39.85$, $SD_{\text{age}} = 13.07$) offered further causal support for our proposed effect employing fully randomized designs (meta-analytic effect = 0.39, 95% CI [0.28, 0.49], corresponding to an approximate 16% decrease after the policy implementation; see Fig. 2 and *SI Appendix* for details). Together, these findings—both from real-world data with high external validity and experimental data with high internal validity—provide evidence of the robustness and generalizability of the effect.

Furthermore, our experimental studies provided evidence that this effect can be attributed to perceptions of personal losses vs. societal gains associated with system-level policies being greater before than after their implementation. Our research thus contributes to the existing body of evidence by positing that public opposition to system-level policies not only represents preferences against the policy (10) but is also, in part, a psychologically transient phenomenon that decreases following the policy's implementation. For instance, the current body of evidence, primarily based on correlational analyses, shows that the introduction of some system-level policies in the health domain is associated with an increase in public support for these policies (7); or that newspaper coverage sentiment on smoking bans and carbon taxes becomes

more positive from the announcement of these system-policies to their implementation and afterward (14, 44). Our work provides causal evidence for this effect, demonstrates its prevalence across various intervention contexts, and elucidates its underlying psychological mechanisms. While initial negative reactions to such government policies might be intense, they tend to weaken once the policy is enacted.

The findings also enhance our understanding of psychological reactance more generally. According to reactance theory, policies that may eliminate or have eliminated individual choices and freedoms create a state of reactance (5). Prior research has found that reactance depends on different factors, such as culture or personality (8, 45, 46). We add to this by documenting that psychological reactance is time-dependent; that is, psychological reactance depends on whether a threat to individual choices or freedoms will occur in the future or occurred in the past.

We rely on a transition heuristic account to explain why individuals may prioritize personal losses (i.e., the individual level) over societal gains (i.e., the societal level) before vs. after the implementation of system-level policies, thereby shedding light on the temporal dynamics of reactance. However, it is reasonable that psychological reactance to system-level policies is determined by multiple psychological factors, and we do not claim that our proposed mechanism is the sole determinant. In fact, our current work echoes previous calls for a systematic research program on this important phenomenon (1, 14). Such a research program could be structured along three major themes: i) correlates and consequences of psychological reactance to system-level policies; ii) (alternative) explanations of temporal dynamics in psychological reactance to system-level policies; and iii) interventions to increase the public acceptance of system-level policies.

First, future work should further investigate the link between psychological reactance and other related cognitive and behavioral outcomes of system-level policies. For instance, how do policy acceptance, trust in governments, or social norm activation change from the announcement of a system-level policy to its implementation and beyond? To provide initial insights into such questions, we conducted an additional study investigating the impact of an ex ante vs. ex post system-level policy implementation on injunctive and descriptive norm perceptions ($N = 261$; context: tax increase on alcohol as in Study 2; see Study S2 reported in *SI Appendix*). We find a marginally significant decrease in injunctive norm perceptions after the system-level policy's implementation ($P = 0.083$), but no significant decrease in descriptive norm perceptions ($P = 0.906$). While these findings suggest that the implementation of system-level policies may not immediately affect social norm activation, future research is needed to investigate their medium- to long-term effects. Moreover, future research might more directly investigate psychological reactance in response to system-level policies that lead to restrictions for oneself vs. for others. Research in the context of mandatory vaccinations shows that reactance is smaller when the mandate does not affect oneself (e.g., ref. 10). We interpret this finding as additional support for our transition heuristic account as the underlying reasoning, particularly on personal losses, should only matter for policies that influence/restrict oneself.

Second, future work should explore other explanations for the observed effect that might relate to or be independent of our proposed transition heuristic account. For instance, psychological reactance to a system-level policy might diminish after its implementation because individuals adapt emotionally to the policy (18). Moreover, it is also possible that people view these policies as definite and unchangeable, and therefore see them in a more positive way, a tendency referred to as rationalization (40, 47).

While we acknowledge that these related mechanisms could also play a role in explaining the documented effects, our research offers at least some empirical evidence for our proposed process through mediation and moderation. Notably, our effects remain robust even when seeking to empirically account for related mechanisms, specifically by controlling for the degree to which participants believe they can change the policy and how well they think they can emotionally adapt to it (*SI Appendix, Study S1*). In fact, our proposed process may provide insights into why individuals may emotionally adapt to or rationalize system-level policies after their implementation. Our work shows that individuals shift their focus from personal losses to societal gains when a system-level policy is implemented vs. planned. This shift may, in turn, serve as an important mechanism in why emotional and cognitive reactions to the policy change after its implementation. For instance, increased attention to policy-induced societal gains may facilitate the conscious justification of systemic change, shedding light on the proposed—albeit yet not fully understood—active rationalization processes postulated in system identification theory (47).

Third, a future research program should expand on the implications of our findings for policymaking. For instance, how can policymakers increase public acceptance of system-level policies in response to pressing societal challenges? We found that reactance to planned system-level policies decreases when their benefits for society at large are made more salient (e.g., by communicating their societal benefits). Decision-makers thus could try to reduce reactance to planned system-level policies by interventions that shift people's attention from their personal losses to societal gains. This reasoning aligns with previous research showing, for example, that communication about herd immunity through vaccination reduces reactance to mandatory vaccination (43). Our work demonstrates that this intervention strategy can also be applied to other policy contexts and, more importantly, provides a theoretical explanation for its effectiveness.

Future work could investigate how the acceptability of system-level policies might depend on their choice architecture. For example, future work might investigate how opposition to system-level policies depends on whether the public has a choice to “opt-out” from some features of the policy (e.g., the public could vote on whether some features of the policy should not be part of the law), or whether the public can change back the system-level policy after a certain amount of time. Previous research also demonstrates that states are more likely to abide by international treaties when the treaty design includes opt-out rather than “opt-in” clauses (48). Based on this, we speculate that the implementation and acceptability of system-level policies may be higher when they are implemented at the federal level (e.g., the US federal government or the European Union) but include opt-out clauses at the local or state level (e.g., the US state government or individual European countries). To increase the practical value of such a research program, researchers should collaborate with policymakers to systematically document and analyze public acceptance of and reactance to system-level policies. Similar to “nudge units” that offer evidence-based recommendations for individual-level policies, such initiatives could synthesize knowledge on reactions to system-level policies, educate the public and decision-makers, and influence public thinking. These programs could collect data on past, planned, and ongoing implementations of system-level policies, relying on various data sources such as surveys, experiments, and online or media data. For example, previous research documented that public opinion changes from the announcement of a new policy to its implementation and afterward, using

sentiment analysis of newspaper coverage (14, 44). One may expand such analyses by collecting online data, employing automated sentiment analysis tools (49), or by examining whether news articles focus more on the individual level (e.g., personal losses) over the societal level (e.g., societal gains) before vs. after the implementation of system-level policies. Media coverage can both reflect and drive public opinion (50), and thus may in part contribute to lower acceptability of system-level policies before their implementation.

Limitations. Naturally, our research comes with some limitations. While our observational study using secondary survey data and our experimental studies produced consistent and broadly similar estimates, important differences remain. The observational study included a large set of meaningful control variables (e.g., smoking-related behaviors, sociodemographic variables, and political orientation); yet, unobserved contextual factors (e.g., satisfaction with the incumbent government) may still account for variations in effect sizes across the treated countries (e.g., Belgium and Scotland). In contrast, the experimental studies, while holding such contextual factors constant, may not fully capture real-world processes such as emotional adaptation, which could mitigate psychological reactance post-policy implementation. Future research could address these limitations by systematically investigating factors that shape the magnitude and variability of effect sizes associated with the implementation of system-level policies. Across all our studies, we investigated reactions to system-level policies in the year before vs. in the year after their implementation. Longitudinal studies that track individuals' reactions over time could provide further insights into the long-term dynamics of psychological reactance to system-level policies. Additionally, our studies were conducted with participants from various Western countries; it is unclear to what extent our findings can be generalized to countries that vary in terms of cultural dimensions such as collectivism/individualism or power distance (51, 52). Future research might therefore try to assess how cultural differences in attitudes toward system-level policies and individual freedoms could influence the magnitude and dynamics of psychological reactance. For example, it may be conceivable that psychological reactance to system-level policies may be weaker in societies that put greater emphasis on collective goals (53).

Conclusions. Despite these limitations, our findings highlight the complex nature of public reactions to system-level policies aimed at achieving socially desirable outcomes. When policymakers consider introducing such policies, they (naturally) anticipate that the public might respond negatively to the notion of change. Our results suggest that policymakers should also consider that such responses not only reflect the public's preferences but also represent a transient psychological phenomenon that likely diminishes once the policy is implemented. This broadened decision calculus may be particularly important in times when policymakers need to find solutions to pressing societal problems, such as climate change and pandemics.

Materials and Methods

Ethics and Open Science Statement. Our studies included human participants and were conducted in accordance with the guidelines of the German Psychological Society (DGPs). Participants provided informed consent. There were no physical, psychological, or social risks to the participants. The participants were not deceived. The series of experiments was considered negligible risk and received ethical approval by the Departmental Review Board of the Department of Occupational, Economic, and Social Psychology at the University of Vienna (Approval No. 2025/W/002).

All experimental studies were preregistered. The preregistration protocols, datasets, analysis code, and study materials are openly available through the Open Science Framework (<https://osf.io/6qajnl/>).

Study 1. Study 1 ($N = 49,674$; $M_{\text{age}} = 47.81$, $SD_{\text{age}} = 18.43$; 57% female) relied on data from two representative surveys conducted in the European Union in September–October 2005 and October–November 2006 (27, 28). Between these data collection waves, three countries implemented a smoking ban in indoor workplaces. Belgium banned smoking in all workplaces in January 2006, Scotland banned smoking in all workplaces and public places such as hospitality venues in March 2006, and Spain banned smoking in all workplaces in January 2006 but allowed for some exceptions in hospitality venues. Specifically, Spain's partial smoking ban allowed hospitality venues to establish separate smoking areas, and hospitality venues smaller than 100 m² could choose whether to be smoke-free or not (29). All other EU countries experienced no change in smoking policy: France, the Netherlands, Italy, Luxembourg, Denmark, Ireland, Greece, Portugal, Finland, Sweden, Austria, Cyprus (Republic), Czech Republic, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Slovakia, Slovenia, Germany, and Great Britain (except Scotland). In both data collection waves, public opposition to a smoking ban in indoor workplaces was measured using the same question ("Are you in favor of smoking bans in offices, and other indoor workplaces?"; 1 = Totally in favor, 2 = Somewhat in favor, 3 = Somewhat opposed, 4 = Totally opposed). This allowed us to compare public opposition to a smoking ban in indoor workplaces (our dependent variable) before and after its implementation between countries that implemented the intervention (intervention group) and those that did not (control group). Furthermore, both data collection waves measured a broad set of variables related to smoking behavior, as well as political and sociodemographic characteristics (see *SI Appendix* for wording), which we included as control variables in our analyses.

Study 2. We recruited UK participants ($N = 719$; $M_{\text{age}} = 40.09$, $SD_{\text{age}} = 13.12$; 74% female) through the online panel provider Prolific (<https://www.prolific.com/>) and randomly assigned them to one of six conditions in a 2 (policy implementation stage: ex ante vs. ex post) $\times$ 3 (policy context: cars vs. alcohol vs. meat) between-participants design. The study was programmed with Qualtrics and was preregistered (https://aspredicted.org/DD7_L3Z). We preregistered the hypothesis that psychological reactance (as measured via people's anger toward the policy) is greater ex ante vs. ex post the system-level policy implementation.

Participants first read a short preamble related to the policy context. For example, in the car policy context condition, they were asked to imagine that they sometimes like to drive their car to work or take trips at the weekend (see materials in *SI Appendix*). In the ex ante (vs. ex post) condition, participants read that the government plans to increase (vs. increased) taxes by 30% in the future (vs. past). They further read that their government is planning to increase (vs. increased) taxes by 30% in 1 y from now (vs. 1 y ago). In the cars (vs. alcohol vs. meat) policy context condition, participants read that to reduce carbon emissions (vs. to make people live healthier lives vs. to make people eat more vegan products), their government plans to increase/increased taxes on cars (vs. alcohol vs. meat) by 30% in the future/past. Next, participants were asked to imagine that in a year's time (vs. a year has passed since) their government plans to increase (vs. has increased) taxes on cars (vs. alcohol vs. meat). Following the preamble "If in 1 y's time (vs. after a year has passed since) my government will increase (vs. increased) taxes on cars/alcohol/meat by 30% to reduce carbon emissions/to make people live healthier lives/to make people eat more vegan products, then,;" we measured participants' reactance to the government intervention by using the following item: "I would feel great anger towards the government" (1 = Not at all, 6 = Very much).

Studies 3A and 3B. Participants in both studies were randomly assigned to one of two conditions (policy implementation stage: ex ante vs. ex post) in a between-participants design. Both studies were programmed with Qualtrics and preregistered (Study 3A: https://aspredicted.org/9C2_ZG4; Study 3B: https://aspredicted.org/257_GBN). In both studies, we preregistered the hypothesis that psychological reactance (as measured via people's anger toward the policy) is greater ex ante vs. ex post the system-level policy implementation.

In Study 3A, we aimed to recruit 240 UK participants through Prolific and obtained 244 ($M_{\text{age}} = 39.56$, $SD_{\text{age}} = 13.46$; 71% female). Participants were first

asked to indicate their general attitude toward vaccinations ("In general, I have a positive attitude towards vaccinations"; 0 = Very much disagree, 100 = Very much agree). All participants were then presented with a fictitious disease that was currently spreading in many geographical areas and read that a vaccine was available that had been shown to effectively protect against the disease with only rare occurrences of side effects. In the ex ante (vs. ex post) condition, participants read that in order to limit the spread of this new disease, their government plans to mandate (vs. mandated) vaccination for the general adult population in the future (vs. past). They were also told that their government is planning to force (forced) them to get vaccinated in 1 y from now (vs. 1 y ago). Next, participants were asked to imagine that in a year's time (vs. a year has passed since) their government plans to introduce (vs. introduced) mandatory vaccination. Following the preamble "If in 1 y's time (vs. after a year has passed since) my government will introduce (vs. introduced) mandatory vaccination to limit the spread of the new disease, then,;" we measured participants' reactance to the government intervention using the following item "I would feel great anger towards the government" (1 = Not at all, 6 = Very much).

In Study 3B, we aimed to recruit 800 German participants from the German online panel provider Clickworker (<https://www.clickworker.de/>) and obtained 809 ($M_{\text{age}} = 39.17$, $SD_{\text{age}} = 12.27$; 42% female). The stimuli were presented in German (see materials in *SI Appendix* for the translated stimuli). The experimental design and procedure were analogous to Study 3A. We first asked participants about their general attitude toward speed limits on German motorways ("I am in favor of a speed limit on German motorways"; 0 = Not at all, 100 = Very much). In the ex ante (vs. ex post) condition, participants then read that to reduce carbon emissions their government plans to introduce (vs. introduced) a speed limit of 120 km/h on German motorways in 1 y from now (vs. 1 y ago). Following the preamble "If in 1 y's time (vs. after a year has passed since) my government will introduce (vs. introduced) a speed limit of 120 km/h on German motorways to reduce carbon emissions, then,;" we measured participants' reactance to the government intervention with the same item as in Study 3A.

Studies 4A, 4B, and 4C. All three studies were programmed with Qualtrics and preregistered (Study 4A: https://aspredicted.org/1B7_CPD; Study 4B: https://aspredicted.org/46X_JN1; Study 4C: https://aspredicted.org/X29_BFL).

In Study 4A, UK participants ($N = 600$; $M_{\text{age}} = 39.81$, $SD_{\text{age}} = 12.62$; 66% female) were recruited through Prolific and randomly assigned to one of two conditions (policy implementation stage: ex ante vs. ex post) in a between-participants design. We preregistered the hypothesis that psychological reactance* to the system-level policy is greater ex ante vs. ex post its implementation, and that we will explore the mediation of this effect by personal loss perceptions of the system-level policy, societal gain perceptions of the system-level policy, and relative focus on personal vs. societal consequences of the system-level policy.

In this study, we used a ban on the use of cars for commuting to work as a policy context. Participants first read a short preamble that asked them to imagine that they work in a part of the city that is accessible by both car and public transportation but that they prefer to use their car. In the ex ante (vs. ex post) condition, participants then read that to reduce carbon emissions their government plans to ban (vs. banned) people from driving to work in this part of the city in the future (vs. past). They further read that their government is planning to forbid (vs. forbade) them to use their car to commute to work in 1 y from now (vs. 1 y ago). Next, participants were asked to imagine that in a year's time (vs. a year has passed since) their government plans to forbid (vs. has forbidden) them to commute to work by car. Following the preamble "If in 1 y's time (vs. after a year has passed since) my government will forbid (vs. forbade) me to use my car to commute to work to reduce carbon emissions, then,;" we measured participants' reactance based on a subscale of the Salzburg State Reactance Scale (54): i) "I would feel great anger towards the government," ii) "I would feel frustrated about the government," and iii) "I would feel that the government restricts my personal freedom" (1 = Not at all, 6 = Very much; Cronbach's $\alpha = 0.93$). Following the same preamble, we measured personal loss perceptions of the system-level policy ("I think about what I personally lose from this decision"; 1 = Strongly disagree, 6 = Strongly agree), societal gain perceptions of the system-level policy

*In the preregistration, we wrongly wrote that the dependent variable would be participant's anger in response to the policy, although this study used psychological (state) reactance as a dependent variable (see below). We otherwise followed the preregistration.

("I think about what the society gains from this decision"; 1 = Strongly disagree, 6 = Strongly agree), and relative focus on personal vs. societal consequences of the system-level policy ("I focus more on how this decision impacts"; 1 = My personal life, 6 = The whole society).

In Study 4B, we aimed to recruit 1,632 UK participants through Prolific and obtained 1,635 ($M_{age} = 39.85$, $SD_{age} = 13.34$; 54% female). We preregistered the hypothesis that psychological reactance to the system-level policy is greater ex ante vs. ex post its implementation, and that this effect will be moderated by loss frame salience. Participants were randomly assigned to one of four conditions in a 2 (policy implementation stage: ex ante vs. ex post) $\times$ 2 (salience: loss frame vs. control) between-participants design. In the control condition, participants followed the same experimental procedure and read the same experimental stimuli as in Study 4A. In the loss frame salience condition, participants were additionally asked to take a perspective that focuses on the losses caused by the policy. They were asked to focus on the negative effects and the negative reactions and developments caused by the system-level policy. Then, participants were asked to briefly describe their thoughts and feelings in a text field. After providing their qualitative responses, we measured participants' reactance using the same items as in Study 4A (Cronbach's $\alpha = 0.91$).

In Study 4C, we aimed to recruit 620 UK participants through Prolific and obtained 622 ($M_{age} = 40.61$, $SD_{age} = 13.57$; 58% female). We preregistered the hypothesis that psychological reactance to the system-level policy will be reduced when the societal benefits of the policy are made salient. Participants

were randomly assigned to one of two conditions (salience: societal benefits vs. control) in a between-participants design. In the control condition, participants followed the same experimental procedure and read the same experimental stimuli as in the ex ante condition of Study 4B. In the societal benefits condition, participants were additionally informed that the government cited benefits for society as a whole as the main reasons for its decision. Participants then read that the policy will, for example, improve public health (e.g., through less air pollution), combat climate change (e.g., through lower carbon emissions), and lead to safer streets for public transport, cycling, and walking (e.g., through fewer accidents and fatal injuries). We measured participants' reactance using the same items as in Study 4A and 4B (Cronbach's $\alpha = 0.93$).

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